

# White paper: notation and terminology across the compendium

2026-07-29 10:33 PDT

**An audit of mathematical notation and controlled vocabulary in the eleven manuscripts under analysis/report/, with the nine verified inconsistencies and a proposed canonical convention.**

## Method and headline finding

Every `report.Rmd` was searched for the symbols carrying the programme’s substantive quantities (outcome, drug indicator, biomarker, moderation parameter, estimand, decay rate, half-life, sample size) and for the labels naming its designs, architectures, and simulation studies. Counts below are literal occurrence counts in the sources. Nothing here rests on inference about intent.

The answer to the question posed is **no**. The notation is locally coherent within each paper and internally correct, but it is not consistent across the compendium. Nine distinct inconsistencies are verified below, of which three are substantive (the same symbol carries different meanings in different papers) and six are cosmetic or organizational (the same object carries different symbols or labels). The substantive three are the drug indicator, the moderation parameter, and the half-life unit. All three can be repaired mechanically without re-running any simulation, and all three should be repaired before the papers circulate together, because each paper cites its neighbors and a referee reading two of them will meet the collision directly.

## The core system, as intended

Underneath the variation there is a single coherent system, most fully realized in papers 01 and 06.

| Object                         | Symbol         | Definition                                                                                                            |
|--------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------|
| Participant, timepoint         | $i, t$         | index over participants and weekly occasions                                                                          |
| Observed outcome               | $Y_{it}$       | symptom score; code column <code>Sx</code>                                                                            |
| Baseline                       | $BL_i$         | participant-specific pre-treatment level                                                                              |
| Response components            | $BR, PB, TV$   | pharmacological, placebo-belief, natural-history; each a modified Gompertz, each a non-negative reduction in symptoms |
| Binary drug state              | $D_{it}$       | 1 on drug, 0 off drug                                                                                                 |
| Continuous exposure            | $D_{bc,it}$    | 1 on drug, $e^{-\lambda t_{sd}}$ off drug; code <code>Dbc</code>                                                      |
| Time since discontinuation     | $t_{sd}$       | weeks since the last on-drug occasion; code <code>tsd</code>                                                          |
| Time on drug                   | $t_{od}$       | cumulative on-drug time; Gompertz input; code <code>tod</code>                                                        |
| Biomarker                      | $B_i$          | participant-level; standardized $b_i = (B_i - \bar{B})/s_B$                                                           |
| Moderation, covariance channel | $c_{bm}$       | correlation between $B$ and $BR$ on drug; Architecture B                                                              |
| Moderation, mean channel       | $\beta_{bm}$   | multiplier on the treatment effect; Architecture A                                                                    |
| Decay rate                     | $\lambda$      | $\ln 2/t_{1/2}$ ; code <code>lambda_cor</code>                                                                        |
| Carryover half-life            | $t_{1/2}$      | code <code>carryover_t1half</code>                                                                                    |
| AR(1) correlation              | $\rho$         | within-factor serial correlation                                                                                      |
| Target estimand                | $\beta_{bm:D}$ | the <b>bm:D</b> interaction coefficient                                                                               |
| Sample size                    | $N$            | participants                                                                                                          |

Terminology is more stable than notation. Four design families recur throughout, with fixed meanings: **OL** (open-label), **OL+BDC** (open-label plus blinded discontinuation), **CO** (traditional two-period crossover), and **Hybrid** (the Hendrickson 2020 design: open-label titration, blinded discontinuation, then a brief crossover). Three data-generating architectures recur with fixed meanings: **A** (direct mean moderation), **B** (MVN differential correlation),

**C** (combined, with independent channel weights  $c_{bm,A}$  and  $c_{bm,B}$ ). The three-component vocabulary (BR, PB, TV) is used identically in every paper that invokes it. ADEMP terminology follows Morris, White, and Crowther (2019) uniformly, and Monte Carlo standard error is abbreviated MCSE throughout.

## The three substantive inconsistencies

**N1. The drug indicator carries two incompatible meanings.** Papers 01, 02, and 06 use  $D_{it}$  for the *binary* on/off state and reserve  $D_{bc}$  or **Dbc** for the *continuous* exponentially decaying exposure. Paper 03 uses  $D_{it}$  for the continuous form, writing  $D_{it} = \exp(-\lambda t_{sd,it})$  off drug and 1 on drug, three times. A reader moving from 01 to 03 encounters the same symbol with the opposite referent, and the distinction is precisely the object of paper 02’s A1-versus-A2 comparison, so the collision sits on the programme’s central axis. Compounding this, three orthographies for the continuous indicator coexist: the code literal **Dbc** (01, 03, 04, 05, 06, 07, 08), the math form  $D_{bc}$  (02, 06, 07, 09), and  $D_{it}$  (03). Papers 06 and 07 use all three within a single document.

**N2.  $c_{bm}$  is used for a correlation and for a regression coefficient.** Paper 01 is careful:  $c_{bm}$  is a correlation, bounded in  $[-1,1]$ , belonging to Architecture B, while  $\beta_{bm}$  is the dimensionless multiplier of Architecture A. Downstream this discipline lapses. Paper 02’s parameter table collapses the row to “ $c_{bm} / \beta_{bm}$ ”. Papers 07, 08, and 09 report Architecture-A runs parameterized as  $c_{bm} \in \{0, 0.45\}$ ; paper 08’s simulation is Architecture A only, yet  $c_{bm}$  appears thirteen times and  $\beta_{bm}$  never. There is a defensible reason for the slippage, namely that the Architecture-A shift is implemented as `c.bm * bm_z * br_sd` on a standardized biomarker scaled by the BR standard deviation, which is exactly the calibration that makes the numeric value 0.45 comparable across architectures. But that justification is stated only in paper 05, and it is stated in code identifiers. Papers that use  $c_{bm}$  for a mean-channel coefficient without repeating the calibration argument are asserting an equivalence they have not established locally.

**N3. Carryover half-life is quoted in incompatible units.** Papers 01, 02, and 07 use weeks, with the canonical grid  $t_{1/2} \in \{0, 0.5, 1.0\}$ . Papers 04 and 05 use days, with 3 days as the baseline and sweeps over  $\{0.5, 1, 3, 5, 7\}$  and up to 14 days. No conversion statement appears in either group. The values are numerically confusable: a “half-life of 1” means one week in paper 02 and one day in paper 04, a sevenfold difference in the quantity that the compendium exists to study. Paper 05’s own text mixes the two, carrying thirteen day-denominated quantities and one week-denominated quantity.

## The six organizational inconsistencies

**N4. Four symbols for one estimand.** The biomarker-by-treatment interaction appears as  $\beta_{bm:D}$  (03, 07, 08, 09),  $\beta_{bm}^{BR}$  and  $\beta_{bm}^{\text{lumped}}$  (06, where the superscript indexes the component, a genuine extension rather than a conflict),  $\beta_3$  (01),  $\theta$  and  $\hat{\theta}$  (02, 10), and the bare code literal **bm:Dbc** in running prose throughout. Paper 10’s use of  $\theta$  is defensible, since its argument is deliberately generic over interaction coefficients, but paper 02 uses  $\theta_{\text{true}}$  for a quantity that papers 07 to 09 call  $\beta_{bm:D}$  with no bridging sentence.

**N5. Outcome notation alternates between mathematics and code.**  $Y_{it}$  appears in 01, 02, 06, 07, and 09; the code identifier **Sx** appears in model formulas in 01, 02, 03, 04, 05, 06, 07, and 09. Papers 08 and 10 set no outcome symbol at all. Mixing the two is acceptable in a reproducibility section; it is not acceptable in a model definition, where papers 02 and 06 both write the fitted model as  $Sx \sim \text{bm} + \text{t} + \text{Dbc} + \text{bm:Dbc}$  immediately after defining  $Y_{it}$ .

**N6. Label collisions across three families.** The letters A, B, C name architectures, but paper 02 simultaneously uses A1, A2, A3 for its three analysis specifications, 192 occurrences of “A2” alone, in a document that also discusses Architecture A. The prefix S names sensitivity blocks S1 to S6 in paper 02 and sweeps S1 to S12 in paper 05, with different referents at every shared index. Simulation studies are labeled “Study A” and “Study B” in paper 06 but “Study 1” through “Study 5” in papers 03 and 09.

**N7. Two sample-size conventions.** Paper 01 reports  $N = 35$  *per randomization path*; papers 04 and 05 report  $N = 35$  *per design*, that is, in total. Paper 11 documents its own instance of this explicitly, noting that its Architecture C grid “uses the earlier driver’s total-sample convention” and that matched runs are pending, so its cells are not readable against paper 01’s tables. This is the one inconsistency the compendium has already caught in writing.

**N8. Design labels alternate between abbreviation and prose.** Papers 01, 02, and 07 use CO, Hybrid, and OL+BDC as abbreviations; papers 04, 05, and 09 predominantly spell out “traditional crossover”, “hybrid”, and “open-label”. Capitalization of “Hybrid” is not settled even within papers: 01 has 21 capitalized against 3 lowercase, 05 has 12 against 128.

**N9. The sign convention is stated only in paper 06.** Paper 06 fixes that the three components are non-negative reductions, so an increase in any component lowers  $Y$ . Consequently effect sizes are negative throughout the compendium ( $-2.0$  nightmares per week in 04,  $\theta_{\text{true}} = -3.6$  in 02,  $\approx -0.23$  bias in 07) while moderation parameters are positive. No other paper states the convention, so a reader of 07 alone cannot tell whether a negative bias estimate means attenuation or inflation.

## Recommendation

Adopt the table in Section 2 as the compendium’s canonical notation and apply it by a single mechanical pass, in this order.

1. **Fix N1 first.** Reserve  $D_{it}$  for the binary state and  $D_{bc,it}$  for the continuous exposure everywhere, retire the bare `Dbc` from mathematical display (keeping it only in code listings and reproducibility sections), and correct paper 03’s three occurrences. This is the only collision that can cause a reader to misread a result.
2. **Fix N3 next.** Convert papers 04 and 05 to weeks, or state a conversion in both groups’ Methods. Weeks is the better target, since it matches the outcome’s measurement interval and the canonical  $\{0, 0.5, 1.0\}$  grid.
3. **Fix N2 by rule.** Use  $c_{bm}$  only for the Architecture-B correlation and  $\beta_{bm}$  only for the Architecture-A multiplier. Papers 07, 08, and 09 relabel to  $\beta_{bm}$ . Add one sentence stating the calibration  $\beta_{bm} \leftrightarrow c_{bm}$  via  $b_i$  and  $\sigma_{BR}$  to any paper that reports both.
4. **Standardize N4 to  $\beta_{bm:D}$ ,** allowing paper 10 to retain  $\theta$  with an explicit bridging sentence, and paper 06 to retain its component superscripts.
5. **Relabel to remove N6.** Rename paper 02’s specifications to **M1, M2, M3** (analysis *methods*), freeing A/B/C for architectures; prefix sweep labels by paper, as **02-S1** and **05-S1**, or rename paper 02’s blocks to B1 to B6; and adopt “Study 1, 2, ...” uniformly.
6. **Settle N5, N7, N8, N9 editorially.** Mathematics in model definitions and code identifiers only in reproducibility sections; state the sample-size convention in every Methods section and resolve paper 11’s pending re-run; use the abbreviations CO, Hybrid, OL, OL+BDC at first definition in every paper and capitalize Hybrid; and state the sign convention in the Methods of every paper that reports a signed quantity.

A single shared file, either a notation section in `analysis/report/README.md` or a `notation.tex` include alongside `sim-preamble.tex`, should hold the canonical table so that later papers inherit it rather than re-deriving it. The project lexicon at `docs/29-nof1-precision-medicine-lexicon.md` already performs this role for prose terminology and is the natural place to cross-reference.

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Rendered on 2026-07-29 at 10:33 PDT. Source: `~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/whitepaper-notation-audit.md`